

ABRAHAM OWUSU ADUSEI

MPhil Candidate, Biomedical Data Science | Computer Vision & Biomedical Signal Processing

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RESEARCH PROFILE

Researcher in biomedical data science with a background in applied mathematics and research interests in computer vision, signal processing, and video-based physiological monitoring. Current research develops a non-contact, appearance-invariant computer vision pipeline for respiratory rate estimation from video, in support of early pediatric pneumonia screening in low-resource settings. Previous research experience includes biomedical machine learning and computational analysis of pediatric health data, including longitudinal growth patterns in children with cleft lip and palate.

RESEARCH INTERESTS

Computer Vision | Signal Processing | Biomedical Image Processing | Video-Based Physiological Monitoring | Spatiotemporal Deep Learning | Machine Learning

EDUCATION

Master of Philosophy: Biomedical Data Science

2025 – Present

National Institute for Mathematical Sciences (NIMS), Kwame Nkrumah University of Science and Technology (KNUST)

CWA: 80.06 / Straight A's across all coursework (Optimization, Machine Learning, Scientific & Technical Computing, Mathematics for Data Science).

- Thesis (in progress): "A Non-Contact, Appearance-Invariant Computer Vision Framework for Point-of-Care Respiratory Rate Estimation: Towards Early Pediatric Pneumonia Screening in Low-Resource Settings."

Bachelor of Science in Mathematics (Applied Mathematics Concentration)

2019 – 2024

Kwame Nkrumah University of Science and Technology (KNUST)

Second Class (Upper Division)

- Thesis: Topological Data Analysis of the Classification and Prediction of Breast Cancer.

SCHOLARSHIP & AWARDS

Data Science for Child Health Now in Ghana (DS-CHANGE) Scholar, Cohort 2.

2025 – Present

Selected as one of six students for the second cohort of the DS-CHANGE program.

PROFESSIONAL & RESEARCH EXPERIENCE

Data Science Intern

Seattle, WA, USA | 2026

Seattle Children's Research Institute (SCRI)/ University of Washington

- Uncovered hidden longitudinal growth trajectories in pediatric patients with Cleft Lip and Palate (CL/P) by applying Principal Component Analysis (PCA) and K-Means clustering to 18-month Weight-for-Age Z-scores (WAZ).
- Engineered dynamic growth-profile clusters to identify clinical sub-populations exhibiting abnormal physical development; presented analytical findings to clinical investigators.
- Conducted a systematic literature review across biomedical databases to define and standardize computable phenotypes for CL/P research.

Teaching and Research Assistant, Department of Mathematics, KNUST

Kumasi, Ghana | 2023 – 2024

- Instructed advanced undergraduate and postgraduate courses, including Real Analysis (Year 2), Integral Equations I (Year 4), and graduate-level Integral Equations II and Methods of Applied Mathematics for MPhil and PhD candidates.
- Designed, administered, and evaluated problem sets, tutorials, and examinations for 200+ students.

- Served as Academic & Peer Mentor across the department and the NIMS-KNUST Data Science Community, guiding junior students in Python programming, scientific computing, and mathematical analysis.

Biostatistics Intern

Ghana | 2021 – 2022

Hawa Memorial Hospital, Osirem (2022); Holy Family Hospital, Techiman (2021)

- Conducted statistical data cleaning, descriptive modeling, and visualization on clinical patient records to support evaluation of healthcare delivery trends.
- Managed patient health records, verified clinical data integrity, and prepared statistical summaries for internal reporting.

MANUSCRIPT

- Co-author, “Predicting Underweight Status in Infants Under 4 Months with Cleft Lip and Palate (CL/P) Using Machine Learning Models” (logistic regression and gradient-boosted classifiers). Manuscript submitted / under review, 2026.

TECHNICAL SKILLS

- Programming & Frameworks: Python (OpenCV, NumPy, SciPy, Pandas, Scikit-learn, PyTorch, TensorFlow), R, MATLAB, SQL, LaTeX, Git, Linux/Bash.
- Core Competencies: Computer vision, video motion tracking, optical flow, signal decomposition, topological data analysis (TDA), numerical optimization, parallel programming.

CONFERENCES & WORKSHOPS

- International Association for Dental, Oral, and Craniofacial Research (IADR) Conference, San Diego, 2026 — Selected Cohort representative / Co-Author Attendee.
- DS-I Africa Conference, Accra — Tool-a-thon Participant, 2025.
- Workshop on Data Science in Child Health Care, 2025.
- Workshop on Data Science in Child Health Care, 2023.
- Workshop on Quantum Computing, 2023.

LEADERSHIP

- Interim Head of Administration / General Secretary, Unity Hall, KNUST (2021 – 2022).
Elected General Secretary and sworn in as Interim Head of Administration for Unity Hall, the largest residential hall in West Africa (~2,000 residents); co-governed residential budget and policy on the 11-member Hall Council alongside 9 faculty members.
- Elected Parliamentary Council Member, 13th KNUST SRC Parliamentary Council (2021 – 2022).
1 of 41 representatives legislating student governance policy for the university student body.
- Executive Leader & Committee Head, Saviour Church of Ghana Students Union, KNUST Chapter (2020 – Present).

CERTIFICATIONS

- Biomedical Responsible Conduct of Research (CITI Program).
- Human Subjects Research (CITI Program).

ACADEMIC REFERENCES

Available upon request.